Sound is Vibration

Objectives:

This lesson is designed for grade levels 1 - 3.
1. To understand that sound is caused by vibration.
2. To understand that sound travels through solids, liquids and gases.
3. To understand that high pitched sounds are caused by more vibrations and low pitched sounds are caused by fewer vibrations.

Materials Needed:

plastic ruler(s), tuning forks of 2 or 3 different pitches, ping-pong ball attached to a string with tape, boom-box-type radio, balloon filled with air, paper plate, tiny pieces of torn paper, baggie filled with water, scissors, several glass bottles of the same size, large container for water, styrofoam cups, wire coat hangers, string (different types), wire cutters, pencils (with wooden ball attached to the eraser end if possible), stopwatch, flower pots of varying sizes attached to rope, long tubes of PVC pipe cut to different lengths for different tones, xylophone, plastic straws, tape.

Strategy:

1. Ask children if anyone knows what a vibration is. Ask children to vibrate their bodies. Write vibrate on board. Put a plastic ruler overhanging the edge of a table; hit the ruler and watch it vibrate. Ask: "Can we see the vibration? Can we hear the vibration? Can we feel it?" Show tuning fork; hit; listen; touch vibrating tuning fork to ping pong ball attached to a string. Observe the effect; ask same questions. Pass tuning forks around so children can experience the sound and feel the vibration; note differences in pitch. Hold inflated balloon in front of boom-box speaker with volume turned up loud. "Can you feel the vibration? Where is it coming from?" Place small paper plate with tiny pieces of paper on it on top of speaker. (Boombox must be turned so that speakers are facing up.) Observe what happens to the papers. Explain that all sounds are caused by something vibrating; sound is vibration, etc.

2. How do vibrations from tuning fork and ruler reach us as sound? The vibrations cause waves to carry sound through the air. (You will perhaps need to discuss waves, depending on age and experience of children.) "We know sound
travels through air. Does it travel through a solid? Put your ear on the desk. Tap the desk. What do you hear?" Try several such examples. "Does sound travel through a liquid?" Get children to talk about experiences they have had such as talking underwater in the bathtub or swimming pool. Can they hear sound
under water? "Try putting a plastic baggie filled with water next to your ear. Have another child speak loudly directly into the other side of the baggie. Can we hear the sound through the water? Does sound travel through a solid, a liquid, and a gas (air)? When was it the loudest?" (through the solid) "Why do you think this may be true?" (The molecules in a solid are more tightly packed, so the vibration can travel without being dispersed as it does in air.)

3. Show a child's xylophone or an autoharp. Strike a short bar, then a long bar and ask children what they heard. Do the same for the lengths of PVC pipe. (Hit the end of the pipes with the palm of your hand.) Make a kazoo out of a plastic straw by flattening one end and cutting it into a triangular shape; blow. As one person blows through the kazoo, another takes a scissors and snips an inch or so off the end. Keep doing this and ask the students to notice

what's happening to the sound as the straw gets shorter. Help them relate the shortness of length to high pitch through a variety of such experiences.

4. Now allow the children to experiment and "play" with all the materials so that they develop strong physical understanding of vibrations in all the ways that have been presented. Have a set of 8 identical bottles and a container of water and have them fill the bottles with varying amounts of water. See if they can match the scale of the bottles to that of the xylophone. Give children the opportunity to make a string phone or wire-hanger chime. (Tie a length of string to each end of a wire coat hanger. Put the other end of each string in your ears, tap the hanger against something like a desk. Listen for the ringing through the string.)

Performance Assessment:

1. Children must: Arrange bottles with varying amounts of water in row according to pitch. Explain the relationship between the amount of water and the pitch. Tell what is vibrating. Blow across or just into the top of each bottle and explain how the relationship between pitch and the amount of water in the bottle changes. Hypothesize as to why this might be.

2. Make a stereo coat hanger as explained above. Tell how you might make one with a higher pitch (shorter wire). Cut hanger with wire cutter to get different lengths and different pitches.

3. Make a straw kazoo. Listen to its pitch. Have your partner cut off an inch or two with a scissors while you are blowing. What happens to the sound? Cut off another inch. What do you notice now?

4. Listen to the sound of each tuning fork. (Strike tuning fork on your shoe then put it to your ear.) Arrange the tuning forks from highest pitch to lowest. What do you notice about the forks when they look this way?

Conclusions:

The shorter the tube (or straw, or xylophone key or amount of water in the bottle) the higher the pitch. The longer the tube, the lower the pitch. Pitch, (high and low) is caused by the frequency, or number, of vibrations per time. More vibrations per time ("bigger" frequency) = higher pitch; fewer vibrations per time ("smaller" frequency) = lower pitch.

References: